IMEN266 Operations Research II

Concept & Proof Companion — Chapters 1–3 (Basic Probability)

[FULL version — all blanks filled]

Fall 2026

How to use this document

This companion *replaces your class notes*: every definition and theorem from the Ch. 1–3 slides is written out here in full, so you can watch and think in class instead of transcribing. It does **not** replace *doing*: for each core proof, (1) work through the GAP-FILL version with pen and paper, (2) check against the FULL version, (3) answer the Checkpoints, and (4) if stuck — or done — use an AI study card. Exams are closed-book; these six proofs and their checkpoints are exactly the kind of reasoning the exams reward.

1 Concept digest

1.1 Probability basics

A **sample space** Ω collects all outcomes of an experiment; an **event** is a subset $A \subseteq \Omega$. A probability measure P satisfies the axioms (i) $P(A) \geq 0$, (ii) $P(\Omega) = 1$, (iii) countable additivity: for *disjoint* $A_1, A_2, \ldots$, $P(\bigcup_i A_i) = \sum_i P(A_i)$. Useful consequences: $P(A^c) = 1 - P(A)$; if $A \subseteq B$ then $P(A) \leq P(B)$; and for any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad \text{(Proof 1).}$$

The general **inclusion–exclusion** law for $E_1, \ldots, E_n$ alternates sums over single events, pairs, triples, . . . (statement on slide p. 4/60).

1.2 Conditional probability and independence

For $P(B) > 0$, $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ — "probability of A inside the world where B happened." Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$ (equivalently $P(A \mid B) = P(A)$). For three or more events, *mutual* independence requires the product rule for *every* sub-collection, not just pairs.

1.3 Reliability warm-up

With independent components, component i working with probability p_i:

$$R_{\text{series}} = \prod_{i=1}^{n} p_i, \qquad R_{\text{parallel}} = 1 - \prod_{i=1}^{n} (1 - p_i).$$

(verified in notebook: CP1 warm-up) Chaining hurts, redundancy helps; Ch. 9 generalizes both via structure functions.

1.4 Law of total probability and Bayes' rule

Let $B_1, \dots, B_n$ **partition** Ω (mutually exclusive, exhaustive, $P(B_i) > 0$). Then for any event A:

$$\textbf{(LTP)} \quad P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i), \qquad \textbf{(Bayes)} \quad P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_i P(A \mid B_i)P(B_i)}.$$

(verified in notebook: CP1 #1–#3) Read Bayes as: *posterior* $\propto$ *prior* $\times$ *likelihood.*

1.5 Random variables and distribution functions

A **random variable** X assigns a number to each outcome. Its **cdf** $F(x) = P(X \le x)$ always satisfies: non-decreasing; $\lim_{x\to-\infty} F = 0$, $\lim_{x\to\infty} F = 1$; right-continuous. *Discrete* X: pmf $p(x) = P(X = x) \ge 0$, $\sum_x p(x) = 1$. *Continuous* X: pdf $f \ge 0$, $\int f = 1$, $P(a < X < b) = \int_a^b f = F(b) - F(a)$, and $P(X = x) = 0$ for every point x.

1.6 Distribution zoo (as used in CP1–CP2)

Name	pmf / pdf	mean	variance	story (class problem)
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	one part: defective?
Binomial(n, p)	$\binom{n}{x} p^x q^{n-x}$	np	npq	#defects in 50 (CP1 #5a)
Geometric(p)	$q^{x-1}p,\ x \ge 1$	$1/p$	q/p^2	trial of 1st defect (#5b)
Neg. Binomial(r, p)	$\binom{x-1}{r-1} p^r q^{x-r}$	r/p	rq/p^2	trial of 3rd defect (#5c)
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	λ	λ	cars per minute (#6)
Hypergeom.(N, K, n)	$\binom{K}{x}\binom{N-K}{n-x}/\binom{N}{n}$	nK/N	—	lathes w/o replacement (#4)
Uniform(a, b)	$1/(b-a)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	—
Exponential(λ)	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	cell lifetime (CP2 #2)
Erlang(k, λ)	$\frac{\lambda^k x^{k-1} e^{-\lambda x}}{(k-1)!}$	k/λ	k/λ^2	message delay (CP2 #3)
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	—

Memorize the *stories*, not just the formulas: Binomial counts successes in a fixed number of trials; Geometric/Neg.-Binomial wait for the 1st/r-th success; Poisson counts events at a rate; Hypergeometric samples *without* replacement; Erlang is k exponential phases in series.

1.7 Expectation

For discrete X: $E[X] = \sum_x x\,p(x)$; continuous: $E[X] = \int x f(x)\,dx$. **LOTUS** (Proof 5): $E[g(X)] = \sum_x g(x)p(x)$ or $\int g(x)f(x)\,dx$ — no need to find the distribution of $g(X)$. **Linearity** (Proof 4): $E[aX + bY] = aE[X] + bE[Y]$ *always*, with no independence assumption. **Variance**: $\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$ (Proof 5); $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$. **Tail-sum formula** (Proof 3): for $X \ge 0$, $E[X] = \int_0^\infty P(X > x)\,dx$ (continuous) and $E[X] = \sum_{k\ge1} P(X \ge k)$ (nonnegative integer valued).

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2 Core proofs

Notation: $q = 1 - p$. Blanks below are the steps *you* must supply in the gap-fill version.

Proof 1 — Inclusion–exclusion for two events

Theorem. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof. Write $A \cup B$ as a union of *disjoint* pieces:

$$A \cup B \;=\; A \cup (B \cap A^c), \qquad B \;=\; (B \cap A) \cup (B \cap A^c).$$

By additivity applied to each disjoint union,

$$P(A \cup B) = P(A) + P(B \cap A^c), \qquad P(B) = P(B \cap A) + P(B \cap A^c).$$

Subtracting the second equation from the first eliminates $P(B \cap A^c)$ and yields $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. □

General case (idea). For $E_1, \ldots, E_n$, apply expectation to the indicator identity $1 - \mathbf{1}_{\cup E_i} = \prod_i (1 - \mathbf{1}_{E_i})$ and expand the product: each k-fold term contributes $(-1)^k P(E_{i_1} \cap \cdots \cap E_{i_k})$, giving the alternating sum on slide p. 4/60.

Checkpoint — answer before moving on.

- Where exactly did the proof use *disjointness*? Point to the line.
- Why does $P(A) + P(B)$ *overcount*, and what does the subtracted term count?
- Sanity check the general formula for $n = 3$ on a Venn diagram.

AI study card — paste one of these into your AI assistant:

- `Quiz me step by step on the proof that P(A∪B)=P(A)+P(B)-P(A∩B). Ask one question at a time; if I'm wrong, give a hint, not the answer.`
- `Give me a concrete example where someone wrongly uses P(A∪B)=P(A)+P(B). Then make me fix it.`

Proof 2 — Law of total probability and Bayes' rule

Theorem. If $B_1, \ldots, B_n$ partition Ω and $P(B_i) > 0$, then $P(A) = \sum_i P(A \mid B_i) P(B_i)$ and $P(B_j \mid A) = P(A \mid B_j) P(B_j) / \sum_i P(A \mid B_i) P(B_i)$.

Proof. Since the B_i are exhaustive, $A = \bigcup_i (A \cap B_i)$, and since the B_i are mutually exclusive these pieces are disjoint. Additivity gives

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i),$$

where the second equality is the multiplication rule $P(A \cap B_i) = P(A \mid B_i) P(B_i)$. For Bayes, start from the definition and apply the multiplication rule to the numerator and LTP to the denominator:

$$P(B_j \mid A) = \frac{P(B_j \cap A)}{P(A)} = \frac{P(A \mid B_j)\, P(B_j)}{\sum_i P(A \mid B_i) P(B_i)}. \qquad \square$$

Checkpoint — answer before moving on.

- In CP1 #2 (cancer test), name B_1, B_2 and A explicitly, and match each factor of Bayes' formula to a number (.95, .04, .004, .996).

- "Posterior odds = prior odds × likelihood ratio." Derive this one-line form from Bayes for two hypotheses.
- What breaks in LTP if the B_i overlap? Give a 3-event example.

AI study card — paste one of these into your AI assistant:

- `Play a skeptical student. I will explain the proof of the law of total probability; interrupt me whenever a step is not justified.`
- `Construct a base-rate-neglect trap problem different from the cancer test, let me solve it, then grade my solution harshly.`

Proof 3 — Tail-sum formulas for expectation

Theorem. (a) If $X \in \{0, 1, 2, \dots\}$, then $E[X] = \sum_{k=1}^{\infty} P(X \geq k)$. (b) If $X \geq 0$ is continuous, then $E[X] = \int_0^{\infty} P(X > x)\, dx$.

Proof of (a).

$$\sum_{k=1}^{\infty} P(X \geq k) = \sum_{k=1}^{\infty}\sum_{j=k}^{\infty} P(X = j) \overset{(*)}{=} \sum_{j=1}^{\infty}\sum_{k=1}^{j} P(X = j) = \sum_{j=1}^{\infty} j\, P(X = j) = E[X],$$

where $(*)$ swaps the order of summation: the pair (k, j) appears exactly when $1 \leq k \leq j$, so each $P(X = j)$ is counted j times. (Nonnegative terms make the swap legitimate.)

Proof of (b). Write $X = \int_0^{\infty} \mathbf{1}\{X > x\}\, dx$ (the integrand is 1 exactly while $x < X$). Taking expectations and interchanging E and $\int$ (Fubini–Tonelli, nonnegative integrand):

$$E[X] = \int_0^{\infty} E[\mathbf{1}\{X > x\}]\, dx = \int_0^{\infty} P(X > x)\, dx. \qquad \square$$

Corollary (used constantly). For $X \sim \text{Geometric}(p)$: $P(X \geq k) = q^{k-1}$, so

$$E[X] = \sum_{k \geq 1} q^{k-1} = \frac{1}{1-q} = \frac{1}{p}.$$

(verified in notebook: CP2 bridge cell: both tail integrals matched $E[X]$ numerically)

Checkpoint — answer before moving on.

- Draw the (k, j) lattice and shade the region summed in $(*)$ both ways.
- Use (b) to get $E[X] = 1/\lambda$ for the exponential in one line.
- Why does the theorem require $X \geq 0$? What goes wrong for a r.v. that takes negative values?

AI study card — paste one of these into your AI assistant:

- `I will prove E[X]=`$\sum_{k\geq1}$`P(X≥k) by swapping a double sum. Check each index bound I state and refuse to accept hand-waving.`
- `Give me three quick exercises that are one-liners with the tail-sum formula but painful without it.`

Proof 4 — Linearity of expectation (no independence needed!)

Theorem. For any two random variables on the same space and constants a, b: $E[aX + bY] = aE[X] + bE[Y]$.

Proof (discrete case). Let $p(x, y)$ be the joint pmf. By LOTUS applied to $g(x, y) = ax + by$:

$$E[aX + bY] = \sum_x \sum_y (ax + by)\, p(x, y) = a \sum_x \sum_y x\, p(x, y) + b \sum_x \sum_y y\, p(x, y).$$

In the first double sum, sum over y first: $\sum_y p(x, y) = p_X(x)$ (the marginal), giving $a \sum_x x\, p_X(x) = a\, E[X]$; symmetrically the second term is $b\, E[Y]$. At no point did we factor $p(x, y)$ — *independence was never used.* □

Checkpoint — answer before moving on.

- Where *would* independence be needed: $E[XY] = E[X]E[Y]$ or $\mathrm{Var}(X + Y) = \mathrm{Var}\, X + \mathrm{Var}\, Y$? Both?
- Use linearity + indicators to get the Binomial mean np in two lines (write $X = \sum_{i=1}^{n} \mathbf{1}_i$).
- CP1 #4: the lathes are sampled *without* replacement (dependent draws!), yet $E[X] = 2 \cdot \frac{4}{7}$. Which theorem just paid off?

AI study card — paste one of these into your AI assistant:

- `Ask me to compute the expected number of fixed points of a random permutation using linearity and indicators; coach me only with questions.`
- `Try to convince me (wrongly) that linearity of expectation requires independence; I have to locate the flaw in your argument.`

Proof 5 — LOTUS and the variance identity

Theorem (LOTUS, discrete). $E[g(X)] = \sum_x g(x)\, p_X(x)$.

Proof. Let $Y = g(X)$ and group outcomes by their y-value:

$$E[Y] = \sum_y y\, P(Y = y) = \sum_y y \sum_{x:\, g(x)=y} p_X(x) = \sum_y \sum_{x:\, g(x)=y} g(x)\, p_X(x) = \sum_x g(x)\, p_X(x),$$

since every x belongs to exactly one group $\{x : g(x) = y\}$. □

Theorem (variance identity). $\mathrm{Var}(X) = E[X^2] - (E[X])^2$.

Proof. Expand with $\mu = E[X]$ and use linearity (Proof 4):

$$E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu\, E[X] + \mu^2 = E[X^2] - \mu^2. \qquad \square$$

(verified in notebook: CP2 #5: $E[X] = 3$, $E[X^2] = 12$, so $\mathrm{Var} = 3$)

Checkpoint — answer before moving on.

- In LOTUS, which side of the identity requires knowing the distribution of $g(X)$, and why is avoiding that a big deal in practice?
- Constants slide out squared: prove $\mathrm{Var}(aX + b) = a^2\, \mathrm{Var}(X)$ in two lines.

- CP2 #5 computed $E[X^2]$ by LOTUS. What integral would you have needed *without* LOTUS?

AI study card — paste one of these into your AI assistant:

- `Generate a pdf for which E[X] exists but E[X^2] does not; make me verify both claims by integration.`
- `I claim Var(X)=E[X^2]-(E[X])^2 can be negative for some X. Argue with me until one of us proves the other wrong.`

Proof 6 — The memoryless property (exam classic)

Theorem. (a) If $X \sim \text{Geometric}(p)$, then for integers $m, n \geq 0$: $P(X > m+n \mid X > m) = P(X > n)$. (b) If $X \sim \text{Exponential}(\lambda)$, then for $s, t \geq 0$: $P(X > s+t \mid X > s) = P(X > t)$.

Proof of (a). First, the survival function: $X > k$ means the first k trials were all failures, so $P(X > k) = q^k$ (no binomial coefficient — there is only one way to fail k times in a row). Then

$$P(X > m+n \mid X > m) = \frac{P(X > m+n,\ X > m)}{P(X > m)} = \frac{P(X > m+n)}{P(X > m)} = \frac{q^{m+n}}{q^m} = q^n = P(X > n),$$

where the second equality holds because $\{X > m+n\} \subseteq \{X > m\}$. $\square$

Proof of (b). With $P(X > t) = e^{-\lambda t}$:

$$P(X > s+t \mid X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t). \qquad \square$$

Remark (uniqueness, starred). These are the *only* memoryless distributions in their classes: memorylessness forces the survival function to satisfy $g(s+t) = g(s)\,g(t)$, whose only monotone solutions are $g(t) = e^{-\lambda t}$ (continuous) and $g(k) = q^k$ (integer).

Checkpoint — answer before moving on.

- Why does $P(X > k) = q^k$ carry no $\binom{k}{\cdot}$ factor, while $P(X = k)$-type binomial counts do?
- "A used exponential component is as good as new." Say precisely what is being conditioned on, and what is claimed.
- Show Erlang$(2, \lambda)$ is *not* memoryless: compare $P(X > 2/\lambda \mid X > 1/\lambda)$ with $P(X > 1/\lambda)$ numerically.

AI study card — paste one of these into your AI assistant:

- `Quiz me one step at a time: derive the memoryless property for the geometric and the exponential. Then make me prove that no other continuous distribution has it.`
- `Claim (falsely) that the Erlang distribution is memoryless because it is built from exponentials. I must refute you with a computation.`

Where to next. Notebook bridge cell (CP2, last section) → these proofs → Checkpoint quiz at the start of the next lecture → HW 1 Part A (closed-book style, no AI on the first attempt).